

Technical Document #836: Deep Learning - Comprehensive Overview

Technical Document #836: Deep Learning - Comprehensive Overview

Long short-term memory (LSTM) networks are a type of RNN that can learn long-term dependencies. They use gating mechanisms to control the flow of information.

Convolutional neural networks (CNNs) are designed to process grid-like data such as images. They use convolutional layers to automatically learn spatial hierarchies of features.

Dropout is a regularization technique where randomly selected neurons are ignored during training. This prevents co-adaptation and improves generalization.

Transformer architecture relies entirely on self-attention mechanisms, dispensing with recurrence and convolutions. It has become the foundation for modern NLP models.

Transfer learning is a technique where a model trained on one task is re-purposed on a second related task. It has revolutionized computer vision and NLP.

Recurrent neural networks (RNNs) are designed to process sequential data. They maintain hidden states that capture information about previous elements in the sequence.

Computer vision applications are becoming ubiquitous, from facial recognition to autonomous driving. Deep learning has enabled breakthroughs in visual understanding.

Digital twins are virtual replicas of physical systems. They enable simulation and optimization without risking real-world resources.

Sustainability and environmental responsibility are becoming key considerations in technology design. Green computing aims to reduce the environmental impact of technology.

Robotics combines mechanical engineering, electrical engineering, and computer science. Modern robots can perform complex tasks in unstructured environments.

Autonomous vehicles use a combination of sensors, AI, and mapping to navigate without human intervention. Self-driving technology is rapidly advancing.

Key Takeaways:

- Transformer architecture relies entirely on self-attention mechanisms, dispensing with recurrence and convolutions.
- Convolutional neural networks (CNNs) are designed to process grid-like data such as images.
- Transfer learning is a technique where a model trained on one task is re-purposed on a second related task.